Arithmetic and complexity

Scholz–Brauer inequality for multiplication chains

Difficulty: extreme **Importance:** interesting to specialist

Status: Partially resolved

Rating rationale: Extreme because the unrestricted inequality is a longstanding frontier of addition-chain theory despite many proved structured cases; specialist importance reflects the multiplication-only powering model.
Topic: multiplication counts for matrix powers

Problem statement

An addition chain for a positive integer n is a sequence $1 = a_0 < a_1 < \cdots < a_r = n$ in which every a_i with $i > 0$ is a sum of two earlier terms, which may coincide. Let $\ell(n)$ be its minimum possible length r. Is

$$\ell(2^n - 1) \leq n + \ell(n) - 1 \qquad (n \geq 1)?$$

The case $n = 1$ uses $\ell(1) = 0$. A chain implements a computation of A^n by matrix multiplications; the question is about this multiplication-only model, not every possible matrix-function algorithm.

References and status

Bläser, 2013, Research Problem 2.9. N. Clift, *A Short Note on Exact Equality for the Scholz–Brauer Conjecture* (2024), distinguishes this inequality from the stronger equality and gives a counterexample to equality. T. Agama, *A Progress on the Scholz Conjecture on Addition Chains*, manuscript dated 10 May 2026, §1, still states the general inequality as open; its result imposes additional conditions on an optimal addition chain. Searches for “Scholz Brauer conjecture proof 2026” and inspection of these sources found no proof or disproof of the unrestricted inequality. **Admitted: no resolution located.**

Status check — 2026-09-10

Added Wallner, The Decompressed Tree Size of k-Ary Chains, §7, published April 2026, which still identifies Scholz–Brauer as an open conjecture. Searches for full proofs found structured-chain and almost-all-integer claims, not the universal result; Agama’s 2024 introduction recalls Brauer’s proved star-optimal subclass. Clift’s counterexample concerns equality, not this upper bound. The 2016 “First Conjecture” paper concerns elementary binary-length bounds, a different statement.